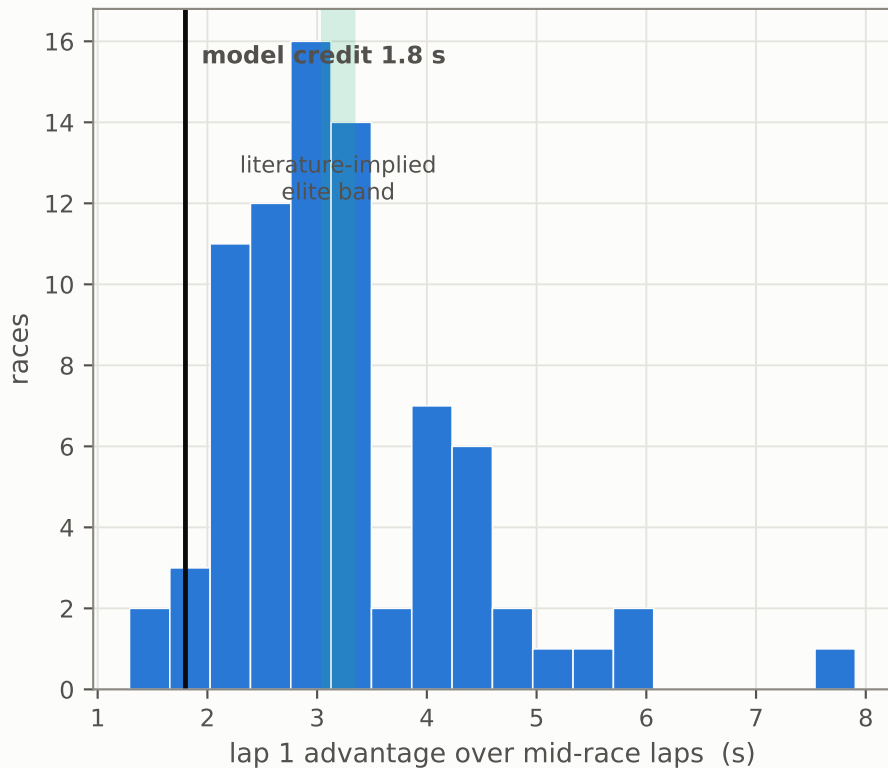
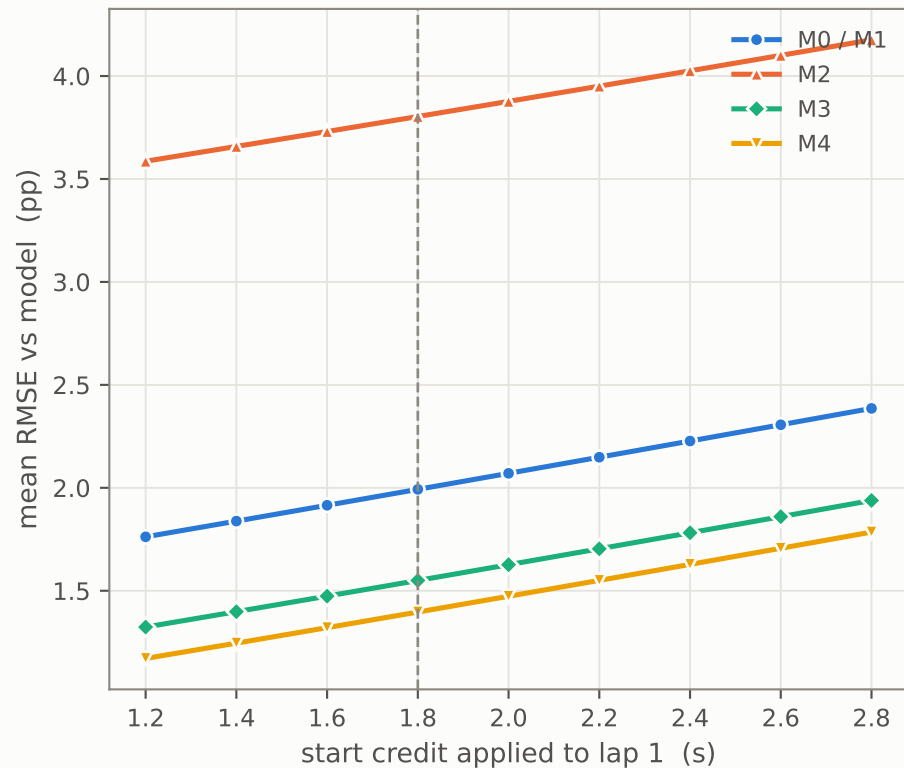


## How much faster the first lap actually is



## Model ranking across the start-credit band



Left: lap 1 is faster than the mid-race laps by far more than the dive alone explains; the excess is pacing and fresh-swimmer effects, which is why the credit must not be estimated from lap differences. Right: the  $M4 < M3 < M0/M1 < M2$  ordering holds across the whole 1.2-2.8 s band.